

6.4.2 System Requirements Analysis Process

The System Requirements Analysis Process in this International Standard is a specialization of the Requirements Analysis Process of ISO/IEC 15288. Users may consider claiming conformance to the ISO/IEC 15288 process instead.

This process **transforms stakeholder requirements** (6.4.1) into **system technical requirements** that guide System Architectural Design (6.4.3). It provides a structured basis for traceability, prioritization, and controlled evolution of requirements throughout the system life cycle.

6.4.2.1 Purpose

The purpose of the System Requirements Analysis Process is to transform defined stakeholder requirements into desired system technical requirements that guide system design.

6.4.2.2 Outcomes

As a result of successful implementation of the System Requirements Analysis Process:

- a) A defined set of system functional and non-functional requirements describing the problem to be solved is developed.
 - b) System requirements are analyzed and trade-offs are performed to optimize the preferred solution.
 - c) System requirements are analyzed for correctness and testability.
 - d) The impact of system requirements on the operating environment is understood.
 - e) System requirements are prioritized, approved, and updated as needed.
 - f) Consistency and traceability are established between system requirements and the stakeholder/customer requirements baseline.
 - g) Changes to the requirements baseline are evaluated for cost, schedule, and technical impact.
 - h) System requirements are communicated to all affected parties and baselined.
-

6.4.2.3 Activities and Tasks

The project shall implement the following activities and tasks in accordance with

applicable organizational policies and procedures for the System Requirements Analysis Process.

6.4.2.3.1 Requirements Specification

This activity defines, analyzes, and documents the system requirements.

- 6.4.2.3.1.1 Analyze the intended use of the system to identify system interfaces, operational and maintenance requirements, design constraints, and qualification requirements.
 - 6.4.2.3.1.2 Develop and document the System Requirements Specification (SyRS) using appropriate modeling and analysis techniques (e.g., use cases, scenarios, data flow diagrams, state models) as applicable.
 - 6.4.2.3.1.3 Perform trade-off analyses to optimize the preferred solution and balance stakeholder priorities, constraints, and technical feasibility.
 - 6.4.2.3.1.4 Analyze and document the impact of the system requirements on the operating environment, including interoperability, performance, and compliance considerations.
 - 6.4.2.3.1.5 Prioritize and approve system requirements based on criticality, feasibility, and stakeholder needs.
 - 6.4.2.3.1.6 Communicate approved system requirements to all affected parties and establish them as the initial baseline.
-

6.4.2.3.2 Requirements Evaluation

This activity evaluates system requirements for quality, feasibility, and alignment with acquisition needs.

- 6.4.2.3.2.1 Evaluate system requirements using the following criteria:
 - Traceability to acquisition and stakeholder needs
 - Consistency with stakeholder baseline
 - Testability and verifiability
 - Feasibility of system architectural design (6.4.3)
 - Feasibility of operation and maintenance
 - 6.4.2.3.2.2 Document the evaluation results, including identified issues, discrepancies, or risks.
 - 6.4.2.3.2.3 Resolve identified issues through clarification, refinement, or change requests to the stakeholder requirements baseline (6.4.1), as appropriate.
-

6.4.2.3.3 Requirements Baseline and Change Evaluation

This activity maintains the integrity of the requirements baseline and evaluates proposed changes.

- 6.4.2.3.3.1 Establish the system requirements baseline once approved and evaluated.
 - 6.4.2.3.3.2 Evaluate proposed updates or changes to the baseline for their impact on cost, schedule, performance, and technical feasibility.
 - 6.4.2.3.3.3 Approve or reject proposed changes in accordance with Configuration Management (6.3.4) and Decision Management (6.3.3) processes.
 - 6.4.2.3.3.4 Communicate approved changes to affected parties and maintain traceability to ensure consistency between updated system requirements and the stakeholder baseline.
-

Cross-References

This process supports and is supported by:

- **6.4.1 Stakeholder Requirements Definition Process** — for initial input requirements.
- **6.4.3 System Architectural Design Process** — for assessing feasibility and guiding design.
- **6.3.3 Decision Management Process** — for evaluating requirement trade-offs and changes.
- **6.3.4 Configuration Management Process** — for maintaining requirement baselines.